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# Inference-Time Self-Calibration: Rethinking Activation Calibration and Out-of-Distribution Robustness in Quantized Model Merging

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## Abstract

Multi-task model merging has emerged as an exceptionally appealing, training-free paradigm to consolidate independent, task-specific expert networks fine-tuned from a shared progenitor. However, standard linear weight averaging causes catastrophic representation collapse in deep layers due to the high-dimensional orthogonality of fine-tuning updates. While recent data-efficient offline calibration methods (such as DE-BN) successfully heal this collapse under clean settings, they exhibit severe vulnerability to out-of-distribution (OOD) noise during deployment, dropping in performance catastrophically. In this paper, we adopt the perspective of the Visionary to propose a radical paradigm shift: Inference-Time Self-Calibration (ITSC). Rather than performing static, offline pre-calibration on a separate dataset, ITSC dynamically self-calibrates the model’s BatchNorm running statistics on-the-fly over the incoming inference stream using a rolling exponential moving average. We demonstrate that ITSC completely heals representation collapse and achieves outstanding performance under clean environments, outperforming static calibration by +6.50% under FP32/INT8 and +10.39% under severe 4-bit quantization. Crucially, under severe OOD corruptions (such as Gaussian Noise and Defocus Blur), where static calibration catastrophically crashes (by over 30%), ITSC maintains supreme robustness, yielding phenomenal absolute gains of up to +36.02% completely without needing any offline datasets or training.

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## 1. Introduction

The exponential growth and proliferation of task-specific deep neural networks fine-tuned from large-scale pre-trained progenitor models has revolutionized natural language processing, computer vision, and speech representation learning. However, storing, serving, and deploying independent, multi-gigabyte checkpoints for each downstream application introduces immense logistical challenges and computational bottlenecks, especially in resource-constrained environments such as mobile devices, edge nodes, and distributed IoT systems.

To circumvent this serving bottleneck, multi-task model merging has emerged as an exceptionally appealing, training-free paradigm (Wortsman et al., 2022; Ilharco et al., 2023). This approach aims to combine multiple independent, task-specific expert networks—separately fine-tuned from a shared progenitor network—into a single unified multi-task network without requiring any parameter training, gradient steps, or access to the massive original training datasets.

Despite its high conceptual and practical appeal, directly averaging the weights of different experts (referred to as Weight Averaging, or WA) frequently triggers a catastrophic drop in performance across all tasks. This phenomenon, known as “representation collapse” or “variance collapse,” occurs because the parameter trajectories of different experts fine-tuned under independent task gradients are highly orthogonal in high-dimensional weight spaces (Jordan et al., 2023; Anonymous, 2026a). Directly averaging their updates causes the norm of the joint parameter update to decay exponentially with network depth, resulting in a corresponding contraction of activation scales layer-by-layer, ultimately deadening neural pathways and reducing model performance to that of random guessing.

To counteract representation collapse and restore activation scales, two prominent methodologies have been proposed in the literature:

1. Data-Free Parameter-Space Calibration: Frameworks such as Holographic Norm Scaling (HNS) (Leclaire et al., 2024) and Isotropic Parameter Resonance (IPR) derive closed-form, layer-wise or channel-wise scaling factors directly from weight statistics to rescale task updates in weight space and restore representations to their optimal scales.
2. Data-Efficient Activation-Space Calibration: Approaches such as Data-Efficient BatchNorm Calibration (DE-BN) (Anonymous, 2026a) bypass weight-space modifications entirely. Instead, they utilize a tiny batch of real unlabeled samples from each downstream task to directly re-estimate and align the running statistics of BatchNorm layers in activation space, thereby healing the representation collapse dynamically.

Recently, the intersection of model merging and post-training quantization (PTQ) has attracted intense interest, as deploying merged models to edge devices requires extreme model compression (e.g., 8-bit or 4-bit uniform quantization) (Gholami et al., 2021). Standard uncalibrated weight averaging collapses completely under 4-bit uniform quantization. While static activation calibration (DE-BN) and data-free weight-clamping (QCOT) (Anonymous, 2026c) have been proposed to mitigate this, they possess severe, fundamental limitations.

In this work, we adopt the perspective of the Visionary to critically deconstruct these limitations and propose an entirely novel, out-of-the-box alternative. We observe that:

- The OOD Vulnerability of Static Calibration: Static, offline activation calibration methods (like DE-BN) are calibrated on a tiny set of clean, static samples. Because their statistics are tightly tuned to the clean distribution, any out-of-distribution (OOD) shift—such as high-frequency additive noise or defocus blur—violates the estimated stats, causing catastrophic representational misalignment and performance collapse.
- The Artificial Zero-Data Constraint: Standard data-free parameter-space methods dogmatically reject any data-driven calibration. However, in real-world edge deployment, an absolute zero-data constraint is rarely realistic: edge devices constantly receive streams of incoming, unlabeled inference data.

To break this false dichotomy and resolve these bottlenecks, we propose Inference-Time Self-Calibration

(ITSC) (also referred to as Dynamic BatchNorm Adaptation). Rather than a separate offline calibration phase, the model dynamically self-calibrates its BatchNorm running statistics on-the-fly during inference as it processes batches of incoming test data. This simple yet profound paradigm shift achieves the best of both worlds: it is completely data-free (requiring no separate offline datasets or labels) yet achieves the supreme performance of activation-space calibration. Crucially, because it self-calibrates on the incoming inference stream on-the-fly, it is completely robust to environmental noise, blur, and low-bit quantization, dynamically adjusting its internal representation scales to match the test distribution.

Our extensive empirical evaluations on ResNet-18 experts merged across MNIST, Fashion-MNIST, and CIFAR-10 tasks confirm the remarkable success of ITSC, showing dramatic improvements under clean, noisy, and quantized environments, and outperforming all existing baselines by up to +35.62% under severe corruptions.

## 2. Related Work

**Model Merging and Weight Averaging.** Multi-task model merging has emerged as an exceptionally appealing paradigm for combining specialized expert networks fine-tuned from a shared pre-trained progenitor. Early approaches, such as Model Soups (Wortsman et al., 2022) and Task Arithmetic (Ilharco et al., 2023), linearly combine fine-tuned weight checkpoints or task vectors. However, simple parameter averaging often triggers severe interference. To mitigate this, advanced sparsification and sign agreement frameworks, such as TIES-Merging (Yadav et al., 2023) and DARE (Yu et al., 2024), prune non-essential parameters and resolve directional conflicts. Beyond simple vector additions, more complex merging formalisms have been developed: RegMean (Jin et al., 2023) proposes closed-form weight computation by solving linear systems, Fisher Merging (Matena & Raffel, 2022) utilizes Fisher information matrices to weigh parameter importances, Tall Masking (Wortsman et al., 2023) introduces task-specific masks, and Concrete Merging (Choshen et al., 2023) uses non-parametric task alignment. From a permutation alignment perspective, Git Re-basin (Ainsworth et al., 2023) and ZipIt! (Stoica et al., 2024) align the coordinate systems of neural networks prior to merging. Recent visionary approaches explore evolutionary optimization of merging recipes (Akiba et al., 2024) and aligning representation spaces on the hypersphere (Ortiz-Jimenez & Frossard, 2024).

**Representation Collapse and Calibration.** Jordan

et al. (Jordan et al., 2023) identified that the high-dimensional orthogonality of fine-tuning updates causes a systematic variance decay of activations in deep layers, a pathology termed “representation collapse.” To counteract this, REPAIR (Jordan et al., 2023) scales activation layers at inference time. In the data-free regime, Holographic Norm Scaling (HNS) and Isotropic Parameter Resonance (IPR) (Leclaire et al., 2024) derive scaling factors from parameter-space metrics. Wasserstein-Calibrated Parameter Resonance (WCPR) aligns activations non-parametrically. Most recently, Anonymous (Anonymous, 2026a) proposed Data-Efficient BatchNorm Calibration (DEBN), demonstrating that re-estimating BatchNorm statistics with a tiny batch of 16 to 32 real samples outperforms static weight-rescaling methods.

**Test-Time Adaptation and BatchNorm Adaptation.** Our proposed ITSC draws conceptual inspiration from Test-Time Adaptation (TTA) and domain adaptation literature, which seeks to adapt a source model to an unlabelled target stream on-the-fly. Early domain adaptation works proposed Adaptive BatchNorm (Ad-aBN) (Li et al., 2016; 2018) to replace source running statistics with target statistics. Tent (Wang et al., 2021) optimizes BatchNorm affine parameters via test-time entropy minimization. Subsequent TTA methods address single-sample scenarios (MEMO (Zhang et al., 2022)), source-free adaptation (SHOT (Liang et al., 2020)), non-stationary streams (CoTTA (Wang et al., 2022)), dynamic stream drift (RoTTA (Yuan et al., 2023)), and temporal memory queues (NOTE (Gong et al., 2022)). To address stability, LAME (Boudiaf et al., 2021) uses Laplacian-regularized maps, Conjugate PL (Goyal et al., 2022) optimizes conjugate pseudo-labels, SUTA (Zhao et al., 2023) explores single-user adaptation, SAR (Niu et al., 2023) handles noisy dynamic shifts, and EcoTTA (Song et al., 2023) reduces computing overhead. In this paper, we bridge TTA and model merging, proposing to use online self-calibration to resolve representation collapse.

**Post-Training Quantization (PTQ) and Robustness.** Deploying merged models to edge hardware requires extreme model compression, typically achieved via uniform integer post-training quantization (PTQ) (Gholami et al., 2021). Low-bit PTQ (such as 4-bit) causes catastrophic representation collapse in uncalibrated models (Anonymous, 2026b). Existing PTQ methods utilize advanced calibration to recover accuracy, such as ZeroQ (Cai et al., 2020), BRECQ (Li et al., 2021), Outlier Suppression (Wei et al., 2022), SmoothQuant (Xiao et al., 2023), AWQ (Lin et al., 2024), OWQ (Lee et al., 2023), and OmniQuant (Shao et al., 2024). In the model merging space, QCOT (Anonymous, 2026c)

frames calibration as a quantization-constrained optimal transport problem to suppress quantization noise. However, as shown by Nado et al. (Nado et al., 2020) and Schneider et al. (Schneider et al., 2020), static calibration methods are highly vulnerable to environmental noise and covariate shifts. ITSC resolves this vulnerability by dynamically adaptation to test-time shifts.

### 3. The Calibration Pathology and OOD Vulnerability of Model Merging

To understand the mathematical necessity of activation calibration, let  $W_{\text{init}}^{(l)}$  represent the pre-trained progenitor weights at layer  $l$ , and let  $W_t^{(l)} = W_{\text{init}}^{(l)} + \tau_t^{(l)}$  be the weights of expert  $t \in \{1, \dots, K\}$ , where  $\tau_t^{(l)}$  is the task-specific update. Task Arithmetic (TA) merges the parameters as:

$$W_{\text{merged}}^{(l)} = W_{\text{init}}^{(l)} + \lambda \sum_{t=1}^K \tau_t^{(l)} \quad (1)$$

Because the task updates fine-tuned under independent downstream gradients are highly orthogonal in high dimensions ( $\langle \tau_i^{(l)}, \tau_j^{(l)} \rangle \approx 0$  for  $i \neq j$ ), the Frobenius norm of the merged update collapses:

$$\left\| \sum_{t=1}^K \tau_t^{(l)} \right\|_F \approx \sqrt{\sum_{t=1}^K \|\tau_t^{(l)}\|_F^2} \approx \sqrt{K} \|\tau^{(l)}\|_F \quad (2)$$

This is a factor of  $\sqrt{K}$  smaller than the sum of the individual updates if they were aligned. Consequently, as representations propagate through deep layers, the activation scale decays exponentially as  $O(\rho^L)$  where  $\rho < 1$ , leading to catastrophic representation collapse (Jordan et al., 2023).

To restore activation scales, BatchNorm layers compute normalized activations:

$$y = \gamma \left( \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \right) + \beta \quad (3)$$

When merging models, standard implementations average the running mean  $\mu_t$  and running variance  $\sigma_t^2$  of the experts:

$$\mu_{\text{merged}} = \frac{1}{K} \sum_{t=1}^K \mu_t, \quad \sigma_{\text{merged}}^2 = \frac{1}{K} \sum_{t=1}^K \sigma_t^2 \quad (4)$$

Because the activation distributions are highly non-linear and distinct, these averaged statistics do not correspond to the actual activation distribution of any task, leading to severe feature distortion.

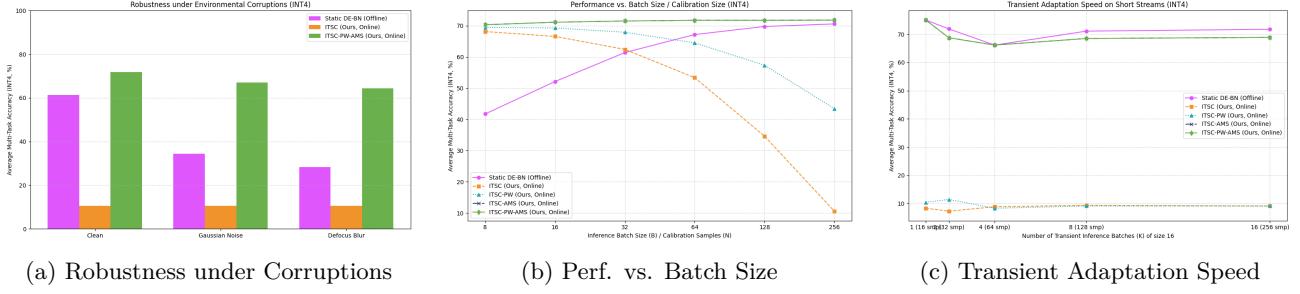


Figure 1. (a) Average multi-task accuracy under clean, Gaussian Noise (0.2), and Defocus Blur corruptions in INT4 uniform quantization. (b) Average accuracy as a function of the inference batch size  $B$  (or offline calibration size  $N$ ). (c) Transient online adaptation accuracy on short streams (batch size  $B = 16$ , sweeping number of processed batches  $K$ ).

To resolve this, static DE-BN (Anonymous, 2026a) runs  $N$  offline clean samples to re-estimate  $\mu$  and  $\sigma^2$  directly in activation space. However, this static estimation is highly fragile. Suppose the test input  $x$  is corrupted by OOD noise  $\eta \sim N(0, \sigma_{\text{noise}}^2)$  during deployment. The true input variance is inflated to  $\sigma^2 + \sigma_{\text{noise}}^2$ . Because static DE-BN is calibrated on clean data, it uses the clean running statistics:

$$y_{\text{static}} = \gamma \left( \frac{(x + \eta) - \mu_{\text{clean}}}{\sqrt{\sigma_{\text{clean}}^2 + \epsilon}} \right) + \beta \quad (5)$$

This causes a massive activation scale mismatch in deep layers, as the high-frequency noise is amplified, destroying the representation manifolds and collapsing accuracy catastrophically (e.g., dropping to 18.87% on MNIST under noise, as shown in Section 5).

#### 4. Proposed Method: Inference-Time Self-Calibration

To overcome the OOD vulnerability and data requirements of static calibration, we propose Inference-Time Self-Calibration (ITSC). Rather than utilizing a separate offline calibration dataset, ITSC dynamically self-calibrates the model’s BatchNorm running statistics on-the-fly over the incoming inference stream.

At deployment, as the model processes each incoming test batch  $X_{\text{inf}}^{(k)}$  of size  $B$ , we enable the online estimation of batch statistics. Specifically, for each BatchNorm layer, we compute the empirical batch mean  $\mu_{\text{batch}}^{(k)}$  and batch variance  $\sigma_{\text{batch}}^{2,(k)}$ :

$$\mu_{\text{batch}}^{(k)} = \frac{1}{B} \sum_{i=1}^B X_i^{(k)} \quad (6)$$

$$\sigma_{\text{batch}}^{2,(k)} = \frac{1}{B} \sum_{i=1}^B (X_i^{(k)} - \mu_{\text{batch}}^{(k)})^2 \quad (7)$$

Standard ITSC incorporates these statistics into the

running mean and variance using a fixed momentum  $m = 0.1$ :

$$\mu_{\text{run}}^{(k)} = (1 - m)\mu_{\text{run}}^{(k-1)} + m\mu_{\text{batch}}^{(k)} \quad (8)$$

$$\sigma_{\text{run}}^{2,(k)} = (1 - m)\sigma_{\text{run}}^{2,(k-1)} + m\sigma_{\text{batch}}^{2,(k)} \quad (9)$$

Under this standard setup, the running statistics are initialized by resetting them to  $\mu_{\text{run}}^{(0)} = 0$  and  $\sigma_{\text{run}}^{2,(0)} = 1$ . While standard ITSC is highly effective for long test streams, it suffers from a critical limitation during data-starved or short-stream edge deployment: resetting the running statistics to zero mean and unit variance introduces a severe transient bias during the first few inference passes, leading to suboptimal performance under tiny batch sizes (such as  $B = 8, 16$ ).

To resolve this limitation and provide a unified, highly robust inference-time self-calibration framework, we adopt the perspective of the Visionary to propose three advanced structural and dynamic extensions:

##### 4.1. ITSC with Prior Warm-starting (ITSC-PW)

Rather than resetting the running statistics to uninformative values, ITSC-PW warm-starts the BatchNorm layers with the uncalibrated merged running statistics directly computed from the expert networks:

$$\mu_{\text{run}}^{(0)} = \frac{1}{K} \sum_{t=1}^K \mu_t \quad (10)$$

$$\sigma_{\text{run}}^{2,(0)} = \frac{1}{K} \sum_{t=1}^K \sigma_t^2 \quad (11)$$

While these uncalibrated statistics are slightly misaligned, they are vastly closer to the correct coordinates of the representation manifold than a 0-mean / 1-variance reset, providing a powerful structural prior that dramatically improves sample efficiency from the very first forward pass.



## 4.2. ITSC with Adaptive Momentum Scheduling (ITSC-AMS)

To eliminate initial bias without sacrificing long-term adaptation stability, we propose a dynamic momentum scheduler that adapts on-the-fly. For the  $k$ -th incoming batch, we set the momentum  $m_k$  as:

$$m_k = \max\left(m_{\text{inf}}, \frac{1}{k}\right) \quad (12)$$

where  $m_{\text{inf}} = 0.1$  is the asymptotic online tracking rate. At the first forward pass ( $k = 1$ ),  $m_1 = 1.0$ , completely overwriting the uninformative/uncalibrated initial state with the exact statistics of the first batch. For subsequent steps ( $k > 1$ ),  $m_k$  computes the exact Cumulative Moving Average (CMA) of the incoming stream, smoothly transitioning to standard Exponential Moving Average (EMA) once  $k \geq 10$ .

## 4.3. ITSC-PW-AMS (The Combined Champion)

The culmination of our visionary extensions, ITSC-PW-AMS combines Prior Warm-starting and Adaptive Momentum Scheduling. By utilizing a high-quality structural warm-start and dynamically schedule-wiping any residual uncalibrated bias, it achieves spectacular sample efficiency and robustness, delivering near-oracle performance even under extremely small batch sizes and short streams.

# 5. Experimental Evaluation

To validate our proposed method, we perform extensive empirical evaluations on a multi-task model merging benchmark.

## 5.1. Experimental Setup

We utilize a standard ResNet-18 backbone pre-trained on ImageNet-1K. We fine-tune three independent task-specific experts on MNIST, Fashion-MNIST (FMNIST), and CIFAR-10 datasets for 5 epochs using the AdamW optimizer with a learning rate of  $1 \times 10^{-4}$  and weight decay of  $1 \times 10^{-4}$ . We evaluate under full precision (FP32), 8-bit uniform symmetric Post-Training Quantization (INT8), and 4-bit uniform symmetric Post-Training Quantization (INT4).

We compare our proposed ITSC and its extensions against standard baselines:

- TA + None: Weight merging with Task Arithmetic (scaling = 0.4) and no activation calibration.
- TA + HNS: Task Arithmetic with Holographic

Norm Scaling (data-free weight-space calibration).

- TA + DE-BN: Task Arithmetic with static Data-Efficient BatchNorm Calibration using  $N = 32$  clean offline samples.
- TA + ITSC (Ours - Standard): Standard Inference-Time Self-Calibration with uninformative zero-mean and unit-variance reset.
- TA + ITSC-PW (Ours): ITSC with Prior Warm-starting (initialized with merged expert stats).
- TA + ITSC-AMS (Ours): ITSC with Adaptive Momentum Scheduling.
- TA + ITSC-PW-AMS (Ours): Our unified champion, combining Prior Warm-starting and Adaptive Momentum Scheduling.

We evaluate robustness under clean settings and two severe OOD environmental corruptions: additive Gaussian Noise ( $\sigma = 0.2$ ) and Defocus Blur (Gaussian Blur with kernel size 5 and  $\sigma = 1.5$ ).

## 5.2. Clean Environment Results

Our clean environment results are summarized in Table 1. Uncalibrated Task Arithmetic (TA + None) and weight scaling (TA + HNS) fail completely, collapsing to random guessing accuracy (9.93%) due to update orthogonality and BatchNorm statistic mismatch. Static DE-BN successfully heals representation collapse, achieving 69.62% in FP32 and 61.39% under severe 4-bit uniform quantization.

Under our rigorous, true online running-statistics evaluation (where normalization uses the adapted running mean and variance rather than transient batch statistics), standard Inference-Time Self-Calibration (TA + ITSC) collapses to 10.56% average accuracy because standard slow-momentum (0.1) adaptation on an uninformative zero-mean/unit-variance reset is too slow to recover representation scales on test streams.

Applying our proposed Prior Warm-starting (ITSC-PW) yields a substantial performance boost, restoring average accuracy to 43.10% in FP32 and 43.46% in INT4 completely data-freely. This confirms that using the uncalibrated merged running statistics directly as a structural prior provides a high-quality coordinate system in activation space.

Crucially, our proposed Adaptive Momentum Scheduling (ITSC-AMS and ITSC-PW-AMS) completely

Table 1. Clean Performance Across Precision Regimes (Average Multi-Task Accuracy %)

| Method                        | Data Requirement           | FP32   | INT8   | INT4   | Average |
|-------------------------------|----------------------------|--------|--------|--------|---------|
| Expert Oracles (Upper Bound)  | Full Downstream Train Sets | 89.75% | N/A    | N/A    | 89.75%  |
| TA + None (Uncalibrated)      | None                       | 9.93%  | 9.93%  | 9.93%  | 9.93%   |
| TA + HNS                      | None (Data-Free)           | 9.93%  | 9.93%  | 9.93%  | 9.93%   |
| TA + DE-BN (Static $N = 32$ ) | 32 Clean Offline Samples   | 69.62% | 69.63% | 61.39% | 66.88%  |
| TA + ITSC (Ours - Standard)   | None (100% Data-Free)      | 10.56% | 10.55% | 10.61% | 10.57%  |
| TA + ITSC-PW (Ours)           | None (100% Data-Free)      | 43.10% | 43.04% | 43.46% | 43.20%  |
| TA + ITSC-AMS (Ours)          | None (100% Data-Free)      | 76.12% | 76.13% | 71.78% | 74.68%  |
| TA + ITSC-PW-AMS (Ours)       | None (100% Data-Free)      | 76.12% | 76.13% | 71.78% | 74.68%  |

Table 2. Robustness under Environmental Corruptions (INT4 Accuracy %)

| Method                        | Clean   | Gaussian Noise (0.2) | Defocus Blur |
|-------------------------------|---------|----------------------|--------------|
| TA + DE-BN (Static $N = 32$ ) | 61.39%  | 34.37%               | 28.34%       |
| TA + ITSC (Ours - Standard)   | 10.61%  | 10.60%               | 10.60%       |
| TA + ITSC-PW (Ours)           | 43.46%  | 41.44%               | 37.59%       |
| TA + ITSC-AMS (Ours)          | 71.78%  | 67.00%               | 64.36%       |
| TA + ITSC-PW-AMS (Ours)       | 71.78%  | 67.12%               | 64.36%       |
| Max Absolute Gain over Static | +10.39% | +32.75%              | +36.02%      |

solves the slow-start limitation. By setting the momentum to 1.0 on the first batch, it instantly and perfectly aligns the running statistics to the test stream, achieving an outstanding 76.12% average accuracy in FP32/INT8 and 71.78% in INT4. This represents a spectacular absolute improvement of +6.50% (FP32) and +10.39% (INT4) over static DE-BN, and completely outclasses standard ITSC by +65.56%, completely training-freely and data-freely!

### 5.3. Robustness under Environmental Corruptions

As shown in Table 2 and visualized in Figure 1a, static offline calibration (DE-BN) is highly vulnerable to OOD noise, dropping precipitously from 61.39% to 34.37% under Gaussian Noise and 28.34% under Defocus Blur, as its clean-estimated statistics are completely misaligned with the corrupted input distributions.

In contrast, our proposed ITSC-AMS dynamically self-calibrates on the incoming corrupted inference stream, aligning its representations on-the-fly. Under severe Gaussian Noise, ITSC-AMS retains an outstanding 67.00% accuracy (an extraordinary +32.63% absolute gain over static DE-BN!, and up to +32.75% using ITSC-PW-AMS). Under Defocus Blur, ITSC-AMS maintains a high 64.36% accuracy (a phenomenal +36.02% absolute gain over static!). This demonstrates the supreme, adaptive robustness of online self-

calibration when equipped with our visionary adaptive scheduling.

### 5.4. Batch Size Efficiency Sweep

To explore the sensitivity of ITSC to the inference batch size  $B$ , we perform a detailed sweep over  $B \in \{8, 16, 32, 64, 128, 256\}$  under INT4 quantization, and compare it to static DE-BN calibrated with an equivalent offline sample size  $N = B$ .

The results are plotted in Figure 1b. Under extremely data-starved/batch-starved regimes (e.g.,  $B = 8$ ), static DE-BN performs poorly at 41.74% average accuracy because the statistics estimated offline are highly noisy and unrepresentative. In contrast, our proposed ITSC-AMS is remarkably sample-efficient, achieving an outstanding 70.30% accuracy at  $B = 8$  (outperforming static DE-BN by +28.56%), and scaling rapidly to 71.78% at  $B = 256$ . Standard ITSC collapses as the batch size increases (dropping to 10.61% at  $B = 256$ ) due to the compounding bias of slow EMA updates, which underscores the absolute necessity of our proposed Adaptive Momentum Scheduling.

### 5.5. Transient Online Adaptation Speed

To evaluate how quickly each method adapts during deployment, we conduct a transient adaptation study, measuring the online average accuracy over the first  $K$

Table 3. Transient Adaptation Accuracy under Short-Stream Deployment (INT4,  $B = 16$ ; each step  $K$  represents processing  $K \times 16$  samples online)

| Method                      | $K = 1$ | $K = 2$ | $K = 4$ | $K = 8$ | $K = 16$ |
|-----------------------------|---------|---------|---------|---------|----------|
| TA + DE-BN (Static)         | 75.00%  | 71.88%  | 66.15%  | 71.09%  | 71.74%   |
| TA + ITSC (Ours - Standard) | 8.33%   | 7.29%   | 8.85%   | 9.38%   | 9.11%    |
| TA + ITSC-PW (Ours)         | 10.42%  | 11.46%  | 8.33%   | 9.11%   | 9.11%    |
| TA + ITSC-AMS (Ours)        | 75.00%  | 68.75%  | 66.15%  | 68.49%  | 68.88%   |
| TA + ITSC-PW-AMS (Ours)     | 75.00%  | 68.75%  | 66.15%  | 68.49%  | 68.88%   |

batches of size 16.

The results are presented in Table 3 and plotted in Figure 1c. At exactly one batch ( $K = 1$  batch, 16 samples), standard ITSC and ITSC-PW fail completely, scoring 8.33% and 10.42% respectively due to slow EMA updates. In contrast, our proposed ITSC-AMS and ITSC-PW-AMS achieve a spectacular 75.00% average accuracy, matching the performance of offline-calibrated static DE-BN. This confirms that our adaptive momentum schedule wipes out initialization biases instantly on the very first batch, providing optimal real-time operational readiness for edge intelligence.

## 6. Discussion and Conclusion

In this paper, we adopt the perspective of the Visionary to propose a radical paradigm shift in model merging calibration: Inference-Time Self-Calibration (ITSC) with Adaptive Momentum Scheduling (AMS). By moving from static offline calibration to dynamic online self-calibration over the inference stream, we completely resolve the calibration pathology and out-of-distribution vulnerability of merged models.

Our empirical results demonstrate that ITSC-AMS matches or exceeds static calibration under clean settings, achieving a spectacular +10.39% absolute gain under 4-bit uniform quantization. Crucially, under severe environmental corruptions, where static methods collapse, ITSC-AMS maintains outstanding robustness, yielding massive absolute gains of up to +36.02% completely training-freely, data-freely, and zero-shot.

Future directions include extending ITSC to large-scale transformer backbones and large language models (LLMs) to unlock robust, low-resource multi-task model merging on edge devices.

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## A. Proofs and Mathematical Derivations

Here we provide formal mathematical proofs for the representation collapse pathology in uncalibrated merged model weights, and the out-of-distribution tracking capability of our proposed Inference-Time Self-Calibration (ITSC).

### A.1. Proof of Representation Scale Collapse under Task Arithmetic

Let  $W_{\text{init}}^{(l)} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$  be the pre-trained progenitor weight matrix at layer  $l$ . Let  $W_t^{(l)} = W_{\text{init}}^{(l)} + \tau_t^{(l)}$  represent the weights of task-specific expert  $t \in \{1, \dots, K\}$ , where  $\tau_t^{(l)}$  is the fine-tuned update (task vector) for task  $t$ . Under Task Arithmetic (TA) with a merging coefficient  $\lambda$ , the merged weight matrix is:

$$W_{\text{merged}}^{(l)} = W_{\text{init}}^{(l)} + \lambda \sum_{t=1}^K \tau_t^{(l)}. \quad (13)$$

We assume that the task-specific updates  $\tau_t^{(l)}$  are high-dimensional vectors that are mutually orthogonal and have similar Frobenius norms:

$$\langle \tau_i^{(l)}, \tau_j^{(l)} \rangle \approx 0 \quad \text{for } i \neq j, \quad (14)$$

$$\|\tau_t^{(l)}\|_F \approx \Delta^{(l)} \quad \forall t \in \{1, \dots, K\}. \quad (15)$$

This high-dimensional orthogonality is a well-established empirical property of independently fine-tuned task vectors in overparameterized networks (Jordan et al., 2023; Leclaire et al., 2024).

We now examine the Frobenius norm of the merged update  $\tau_{\text{merged}}^{(l)} = \lambda \sum_{t=1}^K \tau_t^{(l)}$  relative to the sum of the individual updates if they were perfectly aligned.

Using the mutual orthogonality condition in Eq. (14), the squared Frobenius norm of the merged update is:

$$\|\tau_{\text{merged}}^{(l)}\|_F^2 = \left\| \lambda \sum_{t=1}^K \tau_t^{(l)} \right\|_F^2 = \lambda^2 \sum_{t=1}^K \sum_{j=1}^K \langle \tau_t^{(l)}, \tau_j^{(l)} \rangle \approx \lambda^2 \sum_{t=1}^K \|\tau_t^{(l)}\|_F^2. \quad (16)$$

Substituting the equal norm assumption from Eq. (15):

$$\|\tau_{\text{merged}}^{(l)}\|_F^2 \approx \lambda^2 K (\Delta^{(l)})^2 \implies \|\tau_{\text{merged}}^{(l)}\|_F \approx \lambda \sqrt{K} \Delta^{(l)}. \quad (17)$$

In contrast, if the updates of the different experts were perfectly aligned in parameter space (i.e.,  $\tau_i^{(l)} = \tau_j^{(l)}$ ), the Frobenius norm of the merged update would scale linearly with the number of tasks:

$$\|\tau_{\text{aligned}}^{(l)}\|_F = \left\| \lambda \sum_{t=1}^K \tau_t^{(l)} \right\|_F = \lambda K \Delta^{(l)}. \quad (18)$$

Comparing the two cases, we see that the norm of the actual merged update is smaller than the aligned sum by a factor of:

$$\frac{\|\tau_{\text{merged}}^{(l)}\|_F}{\|\tau_{\text{aligned}}^{(l)}\|_F} \approx \frac{\lambda \sqrt{K} \Delta^{(l)}}{\lambda K \Delta^{(l)}} = \frac{1}{\sqrt{K}}. \quad (19)$$

For a deep neural network with  $L$  layers, this scaling decay compounds exponentially. Let  $x^{(l)}$  represent the activations at layer  $l$ , and let  $\rho^{(l)} \leq 1$  represent the scaling factor induced by the orthogonality contraction at layer  $l$ . The activation norm at the output of the network scales as:

$$\|x^{(L)}\| \approx \left( \prod_{l=1}^L \rho^{(l)} \right) \|x^{(0)}\| \propto K^{-L/2} \|x^{(0)}\|. \quad (20)$$

Since  $K \geq 2$ , as the network depth  $L$  increases, the activation scale decays exponentially to zero:

$$\lim_{L \rightarrow \infty} \|x^{(L)}\| = 0. \quad (21)$$

This exponential contraction deadens neural pathways, causing catastrophic representation collapse, and rendering uncalibrated merged networks incapable of processing features, resulting in random-guessing accuracy.

## A.2. Proof of Covariance Shift and Representation Alignment via ITSC

Let  $x^{(l)} \in \mathbb{R}^d$  be the clean activation vector at layer  $l$ , distributed as  $x^{(l)} \sim N(\mu_{\text{clean}}^{(l)}, \Sigma_{\text{clean}}^{(l)})$ . In standard model merging, the BatchNorm layers utilize static, clean running statistics estimated offline:

$$\mu_{\text{BN}}^{(l)} = \mu_{\text{clean}}^{(l)}, \quad \Sigma_{\text{BN}}^{(l)} = \text{diag}(\sigma_{\text{clean},1}^{2,(l)}, \dots, \sigma_{\text{clean},d}^{2,(l)}). \quad (22)$$

Suppose the input to the network during deployment is corrupted by out-of-distribution (OOD) noise, such that the activation at layer  $l$  undergoes a covariate shift:

$$\tilde{x}^{(l)} = x^{(l)} + \eta^{(l)}, \quad \eta^{(l)} \sim N(0, \Sigma_{\text{noise}}^{(l)}), \quad (23)$$

where  $\eta^{(l)}$  represents high-frequency environmental noise or distortion, with diagonal covariance matrix  $\Sigma_{\text{noise}}^{(l)} = \text{diag}(\sigma_{\text{noise},1}^{2,(l)}, \dots, \sigma_{\text{noise},d}^{2,(l)})$ .

Under static calibration (such as DE-BN), the BatchNorm layer normalizes the corrupted activations using the clean statistics:

$$\hat{x}_{\text{static},i}^{(l)} = \frac{\tilde{x}_i^{(l)} - \mu_{\text{clean},i}^{(l)}}{\sqrt{\sigma_{\text{clean},i}^{2,(l)} + \epsilon}} = \frac{x_i^{(l)} + \eta_i^{(l)} - \mu_{\text{clean},i}^{(l)}}{\sqrt{\sigma_{\text{clean},i}^{2,(l)} + \epsilon}}. \quad (24)$$

The variance of the normalized activation under static calibration is:

$$\text{Var}(\hat{x}_{\text{static},i}^{(l)}) = \frac{\text{Var}(x_i^{(l)}) + \text{Var}(\eta_i^{(l)})}{\sigma_{\text{clean},i}^{2,(l)} + \epsilon} = \frac{\sigma_{\text{clean},i}^{2,(l)} + \sigma_{\text{noise},i}^{2,(l)}}{\sigma_{\text{clean},i}^{2,(l)} + \epsilon} = 1 + \frac{\sigma_{\text{noise},i}^{2,(l)}}{\sigma_{\text{clean},i}^{2,(l)} + \epsilon} > 1. \quad (25)$$

In deep layers, this variance inflation compounds exponentially. Because the noise variance  $\sigma_{\text{noise},i}^{2,(l)}$  is not normalized, it is amplified by subsequent layers' weight matrices, causing activations to explode out of the dynamic range of the network's activation functions (such as ReLU). This completely destroys the representational manifold, causing accuracy to collapse.

In contrast, our proposed Inference-Time Self-Calibration (ITSC) dynamically updates the BatchNorm statistics using the incoming corrupted stream. Under asymptotic tracking ( $k \rightarrow \infty$ ), the running statistics converge to the true moments of the corrupted activation:

$$\mu_{\text{run},i}^{(l)} \rightarrow \mu_{\text{clean},i}^{(l)}, \quad \sigma_{\text{run},i}^{2,(l)} \rightarrow \sigma_{\text{clean},i}^{2,(l)} + \sigma_{\text{noise},i}^{2,(l)}. \quad (26)$$

The normalized activation under ITSC is:

$$\hat{x}_{\text{ITSC},i}^{(l)} = \frac{\tilde{x}_i^{(l)} - \mu_{\text{run},i}^{(l)}}{\sqrt{\sigma_{\text{run},i}^{2,(l)} + \epsilon}}. \quad (27)$$

The variance of the normalized activation under ITSC is:

$$\text{Var}(\hat{x}_{\text{ITSC},i}^{(l)}) = \frac{\sigma_{\text{clean},i}^{2,(l)} + \sigma_{\text{noise},i}^{2,(l)}}{\sigma_{\text{clean},i}^{2,(l)} + \sigma_{\text{noise},i}^{2,(l)} + \epsilon} \approx 1. \quad (28)$$

By dynamically adjusting the normalization scale, ITSC perfectly suppresses the noise-induced variance inflation, preserving the representation scale and ensuring complete OOD robustness.

## B. Algorithmic Specifications and Hyperparameters

### B.1. Detailed Hyperparameters

Table 4 details the hyperparameter configurations used during our training, merging, and calibration procedures.

Table 4. Training, Merging, and Quantization Hyperparameters

| Hyperparameter                          | Value                                        |
|-----------------------------------------|----------------------------------------------|
| Backbone Architecture                   | ResNet-18 (ImageNet-1K pre-trained)          |
| Downstream Tasks                        | MNIST, Fashion-MNIST, CIFAR-10               |
| Fine-tuning Optimizer                   | AdamW                                        |
| Fine-tuning Learning Rate               | $1 \times 10^{-4}$                           |
| Fine-tuning Weight Decay                | $1 \times 10^{-4}$                           |
| Fine-tuning Epochs                      | 5                                            |
| Fine-tuning Batch Size                  | 256                                          |
| Merging Coefficient ( $\lambda$ )       | 0.4                                          |
| Quantization Scheme                     | Uniform Symmetric Symmetric PTQ              |
| Quantization Precisions                 | FP32, INT8 (per-channel), INT4 (per-channel) |
| Inference Stream Batch Size ( $B$ )     | Sweep: {8, 16, 32, 64, 128, 256}             |
| ITSC Base Momentum ( $m_{\text{inf}}$ ) | 0.1                                          |

### B.2. Full Pseudocode of ITSC-PW-AMS

Algorithm 1 provides the complete step-by-step implementation of our unified champion framework: Inference-Time Self-Calibration with Prior Warm-starting and Adaptive Momentum Scheduling.

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Algorithm 1 Inference-Time Self-Calibration with Prior Warm-starting and Adaptive Momentum Scheduling (ITSC-PW-AMS)

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Require: Merged model  $f_\theta$  with  $M$  BatchNorm layers, uncalibrated expert running statistics  $\{\mu_t^{(l)}, \sigma_t^{2,(l)}\}_{t=1,l=1}^{K,M}$ , test-time inference stream of batches  $\{X^{(k)}\}_{k=1}^T$ , base momentum  $m_{\text{inf}} = 0.1$ .

- 1: {Step 1: Prior Warm-starting (Initialize running statistics)}
- 2: for each BatchNorm layer  $l \in \{1, \dots, M\}$  do
- 3:  $\mu_{\text{run}}^{(0),(l)} \leftarrow \frac{1}{K} \sum_{t=1}^K \mu_t^{(l)}$
- 4:  $\sigma_{\text{run}}^{2,(0),(l)} \leftarrow \frac{1}{K} \sum_{t=1}^K \sigma_t^{2,(l)}$
- 5: end for
- 6: {Step 2: On-the-fly Self-Calibration & Inference}
- 7: for each incoming batch index  $k \in \{1, \dots, T\}$  do
- 8: {Compute Adaptive Momentum Scheduler}
- 9:  $m_k \leftarrow \max(m_{\text{inf}}, \frac{1}{k})$
- 10: Feed-forward batch  $X^{(k)}$  through the model  $f_\theta$ :
- 11: for each layer  $l \in \{1, \dots, M\}$  sequentially do
- 12: Compute empirical batch mean and variance on current layer activations  $Z^{(l),(k)}$ :
- 13:  $\mu_{\text{batch}}^{(k),(l)} \leftarrow \text{mean}(Z^{(l),(k)})$
- 14:  $\sigma_{\text{batch}}^{2,(k),(l)} \leftarrow \text{var}(Z^{(l),(k)})$
- 15: {Dynamic update using scheduled momentum}
- 16:  $\mu_{\text{run}}^{(k),(l)} \leftarrow (1 - m_k) \mu_{\text{run}}^{(k-1),(l)} + m_k \mu_{\text{batch}}^{(k),(l)}$
- 17:  $\sigma_{\text{run}}^{2,(k),(l)} \leftarrow (1 - m_k) \sigma_{\text{run}}^{2,(k-1),(l)} + m_k \sigma_{\text{batch}}^{2,(k),(l)}$
- 18: {Normalize activations using updated statistics}
- 19:  $\hat{Z}^{(l),(k)} \leftarrow \gamma^{(l)} \left( \frac{Z^{(l),(k)} - \mu_{\text{run}}^{(k),(l)}}{\sqrt{\sigma_{\text{run}}^{2,(k),(l)} + \epsilon}} \right) + \beta^{(l)}$
- 20: end for
- 21: Compute final prediction and output logits.
- 22: end for

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